
Credit Deterioration Early Warning System

From default prediction to risk monitoring and expected loss

This report explains the current stage of the project. It follows the project from data preparation and model design through validation, probability calibration, expected loss, and the construction of an emerging-risk watchlist. The system is an evolving research prototype, and the closing checklist defines the next development cycle.

Technical project report

10 August 2026

Technical summary

This project builds a credit risk early warning system that reframes default prediction as a time-sensitive deterioration problem. Instead of only classifying loans as default or non-default, the system estimates probability of default over several time horizons, identifies borrowers whose predicted risk is accelerating, and translates those probabilities into expected loss. The goal is an operating view that can place loans into categories such as Healthy, Watchlist, and High Risk while ultimately explaining the borrower-level drivers behind each score. The current analysis establishes that foundation and also makes clear what remains before operational use.

- **The baseline ranking is useful.** A logistic regression trained on loans issued through 2014 and evaluated on resolved 2015 loans achieved a ROC-AUC of 0.7295 and average precision of 0.4067. Because the test default rate was 0.202, average precision was about 2.01 times the random-ranking baseline.
- **Risk bands separate the portfolio clearly, but probabilities are too low in the riskiest bands.** For logistic regression, the observed default rate increased from 5.5% in the lowest-risk quintile to 42.3% in the highest-risk quintile. In that highest quintile, the mean predicted probability was only 34.4%, an underprediction of 7.9 percentage points.
- **Gradient boosting adds little discrimination.** Categorical-aware histogram gradient boosting increased ROC-AUC from 0.7295 to 0.7314 and average precision from 0.4067 to 0.4078. That is a real but operationally small gain. Both models remain under-calibrated at the high-risk end, suggesting that cohort shift and probability calibration matter more than additional nonlinear flexibility.
- **Longer-term loans carry disproportionate risk.** In the resolved 2015 test cohort, 60-month loans represented 34% of exposure but 53% of the modelled expected loss under a constant 60% loss-given-default assumption. Their observed default rate was 36.4%, compared with 14.9% for 36-month loans. This comparison must be read with a selection caveat: only 67.1% of all 2015 60-month loans had a resolved terminal status and entered the lifetime test.
- **Horizon-specific models provide the core early-warning view.** The 12-, 24-, and 36-month models achieved ROC-AUC values of 0.715, 0.701, and 0.698. Their ranking lift was strongest at 12 months, while the cumulative probability paths supported a small “emerging risk” group whose observed 36-month default rate was 24.0%, versus 13.4% under normal monitoring. A new discrete-time hazard benchmark keeps unresolved loans and produces monotone cumulative probabilities by construction, although its event timing remains approximate.
- **The operating layer is now explicit but still scenario-based.** A configurable economic review rule combines predicted loss, a 15% assumed intervention effect, a \$50 review cost, and 20% review capacity. Under those illustrative assumptions it selects 84,149 loans and produces \$40.5 million of scenario net benefit; this is a decision prototype, not a deployment claim.

Bottom line. The current system supports risk ranking, horizon-aware monitoring, and an initial translation from risk to action. It is not finished: the scores should not yet be treated as production probabilities for pricing, capital, or accounting. The next iterations should improve event timing, add dynamic borrower signals, validate probability and state transitions across vintages, and replace illustrative operating assumptions with policy-owned inputs.

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1 The models rank risk well, but understate its level

1.1 The lifetime model provides useful separation

The first modelling stage uses a simple and interpretable benchmark: logistic regression. Its task is not to make a hard yes-or-no prediction at a single threshold. Instead, it assigns every loan a probability of default, which can be used to rank loans, form risk bands, and estimate expected loss.

Table 1. Out-of-time performance on resolved loans issued in 2015.

Model	ROC-AUC	Average precision	AP / default-rate baseline
Logistic regression	0.7295	0.4067	2.01×
Categorical-aware boosting	0.7314	0.4078	2.02×

ROC-AUC answers a ranking question: if one defaulted loan and one non-defaulted loan are selected at random, how often does the model give the defaulted loan a higher score? The value 0.729 means that this ordering is correct about 72.9% of the time. Average precision focuses more directly on the rare positive class. A value of 0.406 against a 0.202 base rate indicates that the score concentrates defaults near the top of the ranking about twice as effectively as random order.

The model's quintiles make that distinction concrete. The highest-risk fifth of the test set defaulted at 42.3%, while the lowest-risk fifth defaulted at 5.5%. The ratio is roughly 7.7 to 1. This is meaningful separation for screening and prioritization.

1.2 Calibration, rather than ranking, is the immediate weakness

Figure 1 compares the average predicted probability in each risk quintile with its observed default rate. Points on the diagonal are perfectly calibrated. Points above it mean that realized defaults exceeded the model's prediction.

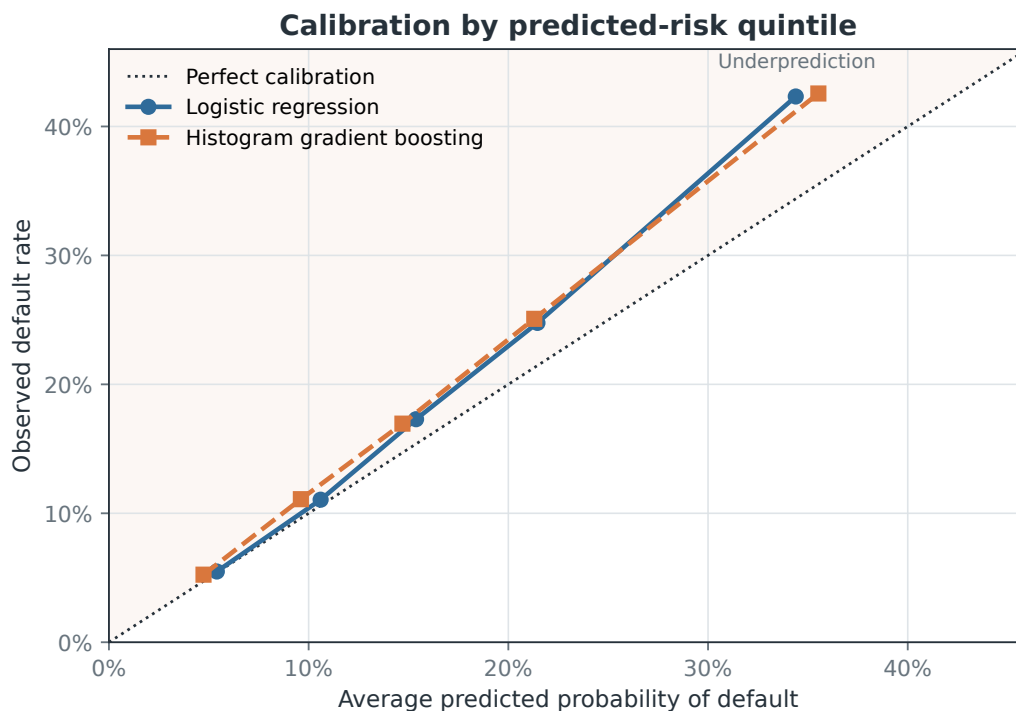


Figure 1. Calibration by predicted-risk quintile for the two lifetime models. Each point represents approximately one fifth of the 375,546-loan test set.

Both curves are close to the diagonal in the lowest-risk bands, then move above it as risk increases. Logistic regression predicts 34.4% in its highest band, where 42.3% is observed. Gradient boosting predicts 35.5%, where 42.6% is observed. The original nonlinear benchmark closes only 0.9 percentage points of the high-band gap.

This distinction matters. A model may rank borrowers well enough to decide whom to review first while still producing probabilities that are too low for pricing or loss estimation. The out-of-time base rate rose from 17.0% in the training cohort to 20.2% in the 2015 test cohort, a shift of 3.16 percentage points. That change is consistent with the observed underprediction and motivates explicit recalibration on a more recent, fully observed validation cohort.

The corrected extension implements this temporal calibration design: the model is fitted through 2013, calibrated on 2014, and evaluated on 2015. Calibration does not change ranking, but it improves Brier score from 0.1474 to 0.1450 and moves mean predicted PD from 15.7% to 17.3%, closer to the observed 20.2%. The remaining gap, especially in the highest decile, shows that recalibration is helpful but not sufficient.

1.3 More flexible classification produces only a marginal gain

Histogram gradient boosting was introduced to capture nonlinear relationships and interactions that an additive log-odds model may miss. In the corrected comparison, nominal categories are explicitly marked as categorical rather than treated as a continuous numeric order. Even so, its improvement is only 0.0019 in ROC-AUC and 0.0010 in average precision. On this feature set, complexity alone does not change the conclusion.

This result should be interpreted carefully. It suggests that the available origination variables already carry most of the ranking information exploited by these two specifications. It does *not* prove that any particular variables are causal drivers, nor that interactions are unimportant. The notebook does not include coefficient analysis, permutation importance, SHAP values, or an ablation study of every individual feature. The new ablation does, however, separate broad information sources: lender-assigned fields alone reach ROC-AUC 0.710, borrower and loan fields alone reach 0.716, and the full corrected model reaches 0.729. This suggests that the system adds information beyond LendingClub’s grade and pricing while still benefiting from those existing underwriting signals.

2 Term, grade, and cohort composition concentrate risk

2.1 Longer terms are riskier, but the resolved-loan sample is selective

The observed default rate is consistently higher for 60-month loans than for 36-month loans in the resolved-outcome data, as shown in Figure 2. The gap is economically important because longer-term loans also tend to remain exposed for longer.

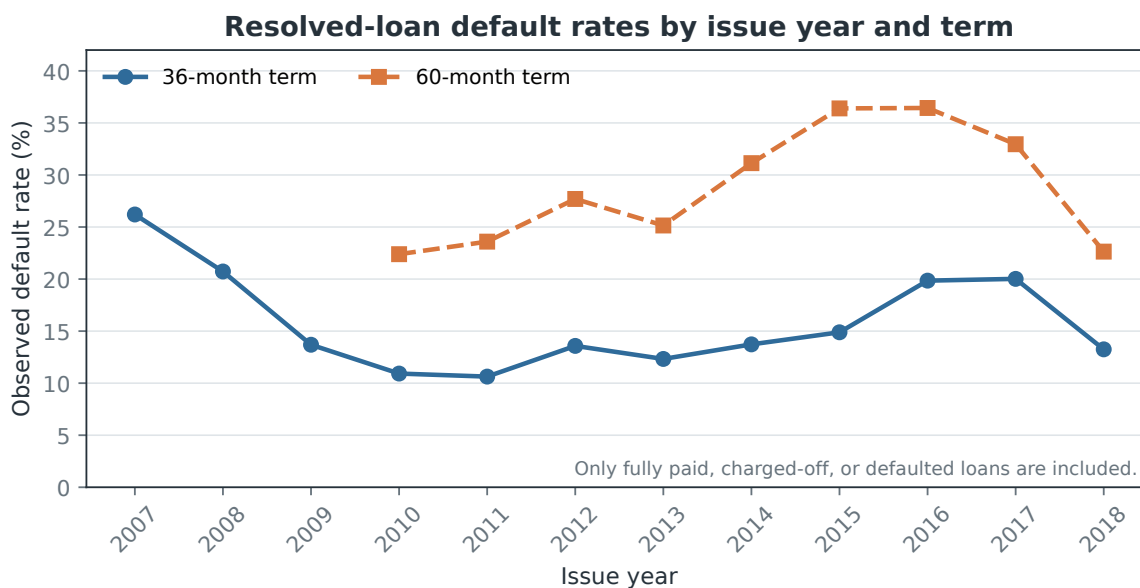


Figure 2. Resolved-loan default rates by issue year and contractual term. The chart describes loans with a terminal good or bad status; it is not a full vintage curve for unresolved loans.

For 2015, the lifetime test contains 283,026 resolved 36-month loans and 92,520 resolved 60-month loans. Within those samples, the model’s ranking is weaker than its overall ROC-AUC: 0.688 for 36-month loans and 0.669 for 60-month loans. Part of the aggregate performance therefore comes from separating the higher-risk term segment from the lower-risk one.

Table 2. 2015 resolved-loan results by term. Expected loss assumes LGD = 60% and EAD equal to original loan amount.

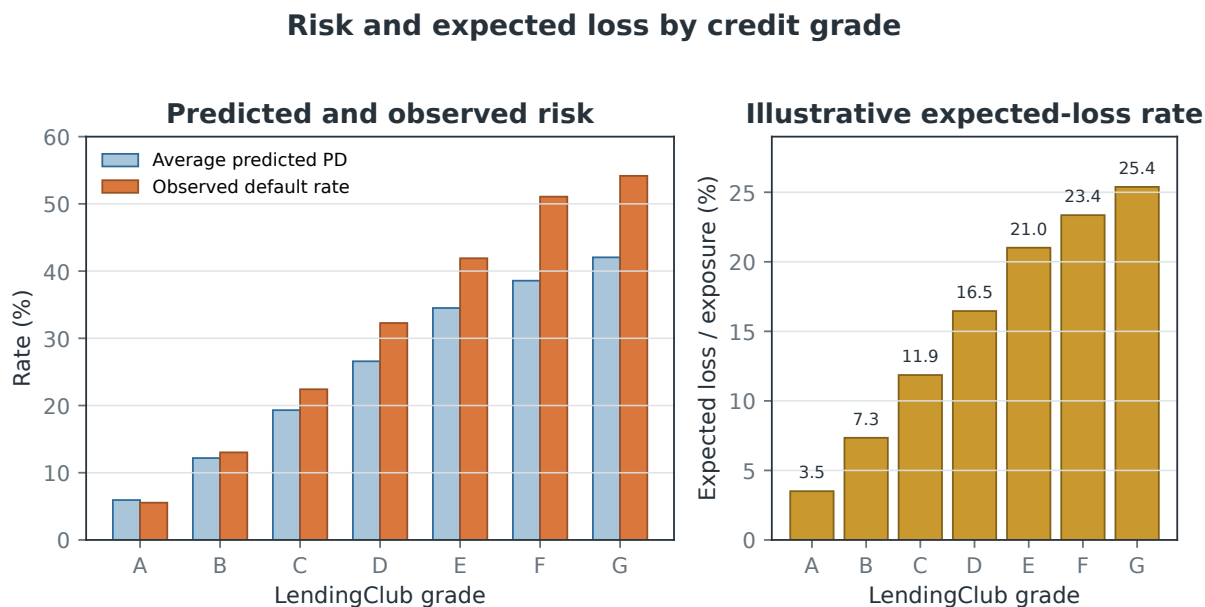
Term	Loans	Exposure	Predicted PD	Observed default	ROC-AUC	Avg. precision	Expected loss	EL rate
36 months	283,026	\$3.624bn	13.63%	14.89%	0.688	0.263	\$285.0m	7.86%
60 months	92,520	\$1.874bn	29.13%	36.39%	0.669	0.515	\$322.4m	17.20%

The exposure comparison is striking: the 60-month segment has about half as much exposure as the 36-month segment, yet slightly more modelled expected loss. That is why term belongs in portfolio segmentation and monitoring.

However, this finding is also where sample construction matters most. Among all 421,095 loans issued in 2015, almost every 36-month loan had resolved by the data cutoff, but 43,227 of the 137,922 60-month loans were still current. Only 67.1% of the 60-month vintage entered the terminal-status test. The 36.4% figure is therefore a default rate among *resolved* 60-month loans, not among all loans issued on that term. It should not be used as an unbiased vintage default estimate.

2.2 Credit grade orders risk, while high grades remain underpredicted

LendingClub grades produce a strong monotonic risk ordering. Figure 3 shows that observed default increases from 5.5% in grade A to 54.2% in grade G. Under the report's illustrative loss assumptions, expected-loss rate rises from 3.5% to 25.4% of original exposure.

**Figure 3.** Predicted probability, observed default, and illustrative expected-loss rate by LendingClub grade for the resolved 2015 test cohort.

The widening gap between predicted and observed risk in grades D through G echoes the quintile calibration result. This is operationally significant: an underestimate in the tail matters more than the same absolute error in a low-risk band because exposure, review intensity, and loss provisions

are often concentrated there.

The grade result also changes how the model should be described. Grade, sub-grade, and interest rate are outputs of LendingClub’s own underwriting process. Including them makes the model useful as a portfolio-risk benchmark built on the lender’s origination decisions, but less suitable as an independent underwriting model. The completed ablation quantifies this distinction: borrower and loan fields alone produce ROC-AUC 0.716, compared with 0.710 for lender-assigned fields alone and 0.729 for the full corrected model.

3 Horizon models reveal when risk emerges

3.1 Near-term risk is easier to rank from origination information

The second modelling stage reframes the target. Rather than asking whether a loan ever ends in default, three models estimate cumulative default risk within 12, 24, and 36 months. The 12-month model has the highest ROC-AUC and the greatest average-precision lift relative to its own default-rate baseline.

Table 3. Out-of-time performance for horizon-specific models on 2015 loans with adequate outcome information for the relevant horizon.

Horizon	Test loans	Default rate	ROC-AUC	Avg. precision	AP lift
12 months	420,801	6.05%	0.7150	0.1364	2.26×
24 months	420,801	13.32%	0.7009	0.2563	1.92×
36 months	420,748	17.37%	0.6982	0.3166	1.82×

Average precision rises with the horizon because the positive class becomes more common. Its lift over the default-rate baseline moves in the opposite direction, from 2.26 times at 12 months to 1.82 times at 36 months. This is a more informative comparison across horizons: origination features distinguish early defaults most strongly, while later defaults are likely to depend increasingly on borrower behaviour, macroeconomic conditions, and other post-origination information.

3.2 Every horizon produces economically distinct risk bands

Figure 4 compares observed default rates across quintiles formed separately within each horizon. Within the 36-month model, observed default rises from 5.3% in the lowest quintile to 33.3% in the highest. The same ordering holds at 12 and 24 months.

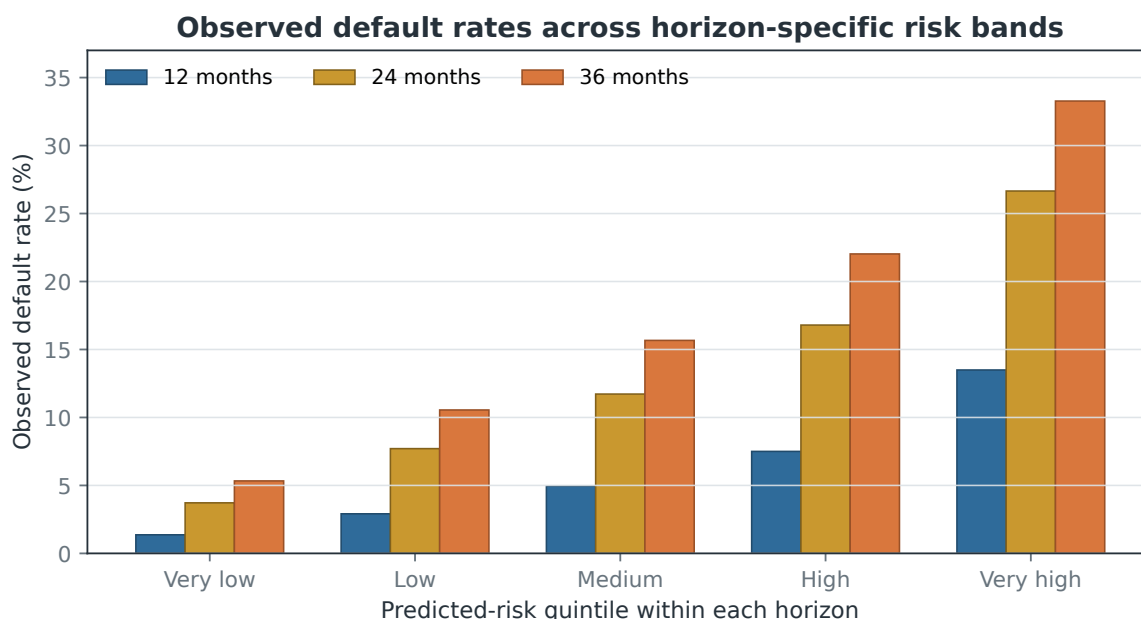


Figure 4. Observed cumulative default rates by horizon-specific predicted-risk quintile. Each band contains about 84,150 loans in the common 2015 cohort.

This result is more useful than a single lifetime score because it separates two questions that often call for different actions: which loans are risky now, and which loans are likely to deteriorate over a longer window? The former can drive immediate review; the latter can drive monitoring frequency, early-warning rules, and scenario analysis.

3.3 Risk acceleration identifies delayed deterioration

The notebook converts the three independent horizon probabilities into monotone cumulative paths. It then defines 36-month risk acceleration as adjusted 36-month PD minus adjusted 12-month PD. Figure 5 shows that loans with larger predicted increases also have markedly higher realized 36-month default.

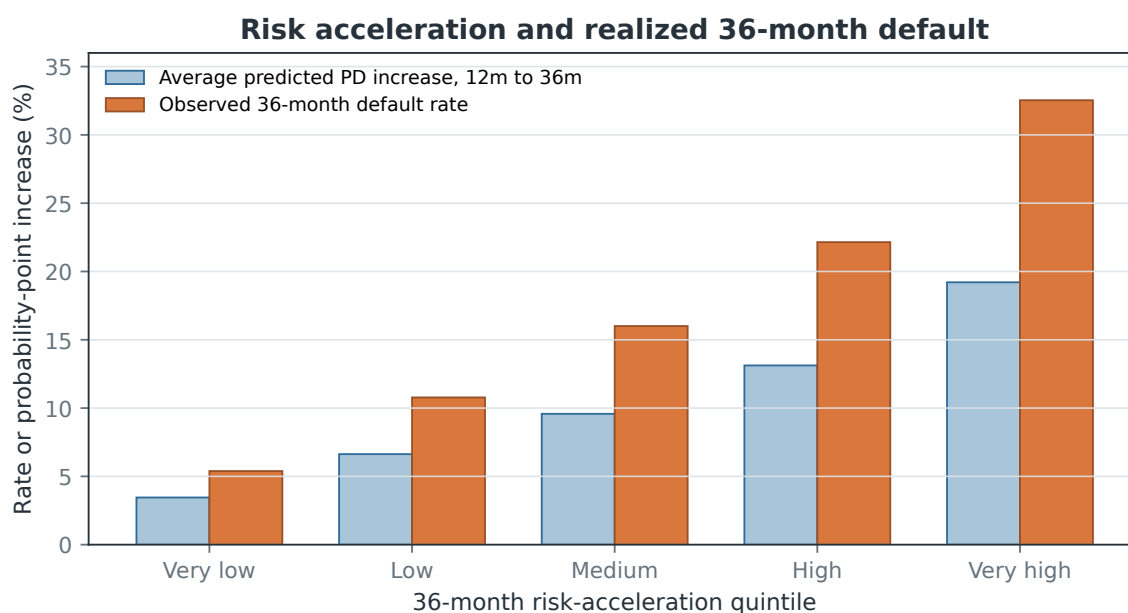


Figure 5. Average predicted increase from 12-month to 36-month PD and observed 36-month default rate by acceleration quintile.

The monotonic pattern gives the acceleration concept empirical support. Observed 36-month default rises from 5.4% in the lowest acceleration quintile to 32.5% in the highest. The statistic is still predictive rather than causal: it is a way of arranging loans by the shape of their modelled risk curve, not evidence that a particular feature causes later deterioration.

4 A horizon-aware watchlist turns scores into an operating view

The final stage groups loans into three monitoring categories:

- **High immediate risk:** 12-month PD is at or above the 80th percentile.
- **Emerging risk:** 12-month PD is below the 60th percentile while the increase from 12-month to 36-month PD is at or above the 80th percentile.
- **Normal monitoring:** all remaining loans.

Table 4. Watchlist profile in the common 2015 horizon cohort. Expected loss uses adjusted 36-month PD, a constant 60% LGD, and original loan amount as EAD.

Group	Loans	Share	Avg. 12m PD	Avg. 36m PD	Actual 12m	Actual 36m	Exposure	36m expected loss
Emerging risk	3,132	0.74%	4.30%	21.11%	6.67%	23.98%	\$57.2m	\$7.3m
High immediate risk	84,150	20.00%	10.71%	28.40%	13.49%	32.67%	\$1.320bn	\$232.4m
Normal monitoring	333,466	79.26%	3.40%	11.89%	4.16%	13.45%	\$5.035bn	\$361.4m

The emerging-risk group is small, only 0.74% of the cohort, but its realized 36-month default rate is almost twice the normal-monitoring rate. Its immediate default rate is much lower than the

high-immediate-risk group, which is exactly the pattern the rule is intended to find: risk that is not extreme at 12 months but becomes material by 36 months.

The table also illustrates a classic portfolio lesson. Normal monitoring contains most loans and therefore accounts for about 60% of total modelled 36-month loss dollars, even though its per-loan risk is lower. A watchlist is useful for review and prioritization, but it does not replace portfolio-wide loss estimation.

These three descriptive categories are still defined by percentile rules, so their meanings can drift as the portfolio changes. The corrected notebook adds a separate economic overlay that ranks loans by expected loss avoided minus review cost and selects positive-value cases within a capacity constraint. Under the illustrative assumptions in the technical summary, the selected 20% of loans have a 29.6% realized 36-month default rate and \$2.005 billion of exposure. The framework is useful because its assumptions are visible and configurable; deployment still requires review capacity, intervention effectiveness, cost, and risk appetite to be estimated and owned by the business.

5 What was modelled and which loans were eligible

5.1 Source population and terminal-status label

The source file, `accepted_2007_to_2018Q4.csv.gz`, contains LendingClub accepted-loan records from 2007 through 2018 Q4. After requiring non-missing loan status and loan amount, the working source contains 2,260,668 rows. The lifetime-default analysis then keeps only loans whose terminal status can be interpreted unambiguously as good or bad.

Table 5. Population flow for the lifetime-default model.

Stage	Loans	Definition
Source after required fields	2,260,668	Non-missing <code>loan_status</code> and <code>loan_amnt</code> .
Resolved outcome sample	1,348,099	Fully paid, charged off, or defaulted, including legacy policy-status variants.
Resolved defaults	269,360	Charged off, default, or legacy charged-off policy status.
Training cohort	453,809	Issue year no later than 2014; default rate 17.03%.
Test cohort	375,546	Issue year 2015; resolved-outcome default rate 20.19%.

The binary target is

$$y_i = \begin{cases} 1, & \text{if loan } i \text{ is charged off or defaulted,} \\ 0, & \text{if loan } i \text{ is fully paid.} \end{cases}$$

Loans marked current, late, or in grace period are excluded from this first stage. That choice gives a clean binary label but creates the resolved-outcome selection described earlier.

5.2 Origination features and preprocessing

The notebook selects 22 source fields, then converts issue date into year and month. Issue year is used to define the temporal split rather than as an intended predictor. The modelled information covers five broad areas:

Table 6. Feature groups used in the project.

Feature group	Variables
Loan structure	Loan amount, contractual term, interest rate, installment, and issue month.
Lender risk labels	Grade and sub-grade.
Borrower capacity	Annual income, employment length, home ownership, verification status, purpose, and debt-to-income ratio.
Credit history	Delinquencies in the past two years, FICO range, inquiries in the past six months, open accounts, public records, revolving balance, revolving utilization, and total accounts.
Time fields for horizon labels	Issue date and last payment date; these are used to form timing labels and are not intended as outcome predictors.

Missing numeric values are replaced with the training median. Missing categorical values are replaced with the most frequent training category. Logistic regression standardizes numeric variables and one-hot encodes categories. Employment length has the largest missing share at 5.83%; other selected fields are nearly complete, with revolving utilization missing in 0.067% and debt-to-income ratio in 0.028%.

All preprocessing is held inside a scikit-learn pipeline, so imputation and encoding are learned on training data and then applied to the test set. This is an important protection against preprocessing leakage.

5.3 Eligibility for the horizon targets

The horizon models require enough follow-up to decide whether a non-default has survived each horizon. The notebook uses last payment date as an approximate timing proxy and includes a row if a default appears within the horizon, if at least that many months are observed, or if the loan was fully paid earlier.

Table 7. Eligible observations and target rates for horizon-specific labels.

Horizon	All eligible rows	Overall default rate	Eligible 2015 test rows
12 months	1,901,296	6.01%	420,801
24 months	1,592,168	13.47%	420,801
36 months	1,422,033	18.18%	420,748

As the horizon lengthens, fewer loans have adequate observation and more have had time to default.

This changing denominator is why performance and default rates must be interpreted separately for each horizon.

6 How the modelling stages were built

6.1 Logistic regression as an interpretable baseline

For a feature vector x_i , logistic regression models the probability of default as

$$\widehat{\text{PD}}_i = \Pr(y_i = 1 \mid x_i) = \sigma(\beta_0 + x_i^\top \beta) = \frac{1}{1 + \exp[-(\beta_0 + x_i^\top \beta)]}.$$

The parameters are fitted by minimizing binary log loss with scikit-learn's default L_2 regularization:

$$\mathcal{L}(\beta) = - \sum_{i=1}^n \left[y_i \log(\widehat{\text{PD}}_i) + (1 - y_i) \log(1 - \widehat{\text{PD}}_i) \right] + \lambda \|\beta\|_2^2.$$

The model is attractive because its output is a probability and its structure is transparent. Its main restriction is additivity in log-odds. Without explicit interaction terms, it cannot easily represent a combination whose joint effect is much stronger than the sum of its parts.

6.2 Histogram gradient boosting as the nonlinear comparison

The comparison model builds an additive sequence of shallow decision trees:

$$f_M(x) = f_0(x) + \sum_{m=1}^M \eta h_m(x), \quad \widehat{\text{PD}}(x) = \sigma(f_M(x)),$$

where h_m is the tree fitted at boosting step m and $\eta = 0.05$ is the learning rate. The notebook uses 200 iterations, at most 31 leaves per tree, at least 100 observations per leaf, L_2 regularization of 0.1, and random seed 42.

Numeric variables receive median imputation. Categorical variables receive most-frequent imputation followed by ordinal encoding, with unseen categories set to -1 . The corrected model passes an explicit categorical-feature mask to the learner, so those codes identify category membership rather than a continuous ordering. This preserves the computational advantage of the histogram learner without imposing an artificial scale on purpose, home ownership, and similar nominal fields.

6.3 Out-of-time validation

The main model comparison is trained on loans issued no later than 2014 and tested on loans issued in 2015. This is more realistic than a random split because it forces the model to score a later vintage. The calibration extension uses a stricter three-way split: fitting through 2013, probability calibration on 2014, and final evaluation on 2015. Both designs reveal temporal drift because the test cohort has a higher resolved-outcome default rate than the earlier cohorts.

The notebook does not tune hyperparameters through cross-validation. The reported results are therefore a direct benchmark comparison, not an optimized model selection exercise. This restraint is useful for interpretation, but uncertainty around the metric differences is not quantified.

7 How performance, timing, and expected loss were measured

7.1 Discrimination and average precision

ROC-AUC can be written as the probability of correct pairwise ordering:

$$\text{AUC} = \Pr(\widehat{\text{PD}}_i > \widehat{\text{PD}}_j \mid y_i = 1, y_j = 0).$$

Average precision summarizes the precision-recall curve:

$$\text{AP} = \sum_n (R_n - R_{n-1}) P_n,$$

where P_n and R_n are precision and recall at successive score thresholds. For a random ranking, expected AP is approximately the positive-class prevalence. This makes $\text{AP}/\Pr(y = 1)$ a useful descriptive lift measure across targets with different default rates.

7.2 Calibration by risk band

For risk band k , the report compares the average predicted probability with the observed default rate:

$$\bar{p}_k = \frac{1}{n_k} \sum_{i \in k} \widehat{\text{PD}}_i, \quad \bar{y}_k = \frac{1}{n_k} \sum_{i \in k} y_i, \quad g_k = \bar{y}_k - \bar{p}_k.$$

A positive gap g_k means that the model underpredicts default in that band. This banded check is easy to communicate, although a production validation should also report Brier score, calibration intercept and slope, confidence intervals, and calibration within material subgroups.

7.3 Horizon labels and monotone probability paths

Let m_i denote the approximate number of months from issue date to last payment date. For horizon $h \in \{12, 24, 36\}$, the target is

$$y_i^{(h)} = \mathbb{1}\{\text{eventual default}_i = 1 \text{ and } m_i \leq h\}.$$

An observation is considered valid for horizon h when

$$V_i^{(h)} = \mathbb{1}\{y_i^{(h)} = 1 \text{ or } m_i \geq h \text{ or fully paid}_i = 1\}.$$

This rule removes non-defaults with insufficient follow-up unless they fully paid early. It is a pragmatic censoring rule, but it is not a formal survival model.

Because the three classifiers are fitted independently, their cumulative probabilities can occasionally violate monotonicity. The notebook applies

$$\tilde{p}_{12} = p_{12}, \quad \tilde{p}_{24} = \max(p_{12}, p_{24}), \quad \tilde{p}_{36} = \max(p_{12}, p_{24}, p_{36}).$$

Violations are rare: $p_{24} < p_{12}$ for 0.018% of the common 2015 cohort and $p_{36} < p_{24}$ for 0.576%. The 36-month acceleration statistic is then

$$\Delta_i^{36} = \tilde{p}_{36,i} - \tilde{p}_{12,i}.$$

The corrected extension also fits a discrete-time hazard benchmark over successive 12-month intervals. If $q_{i,t}$ is the conditional default probability in interval t for a loan still at risk, cumulative probability is

$$\Pr(T_i \leq h) = 1 - \prod_{t \leq h} (1 - q_{i,t}).$$

This construction makes cumulative PD monotone without post-processing and keeps unresolved loans until censoring. The benchmark is trained through 2013, calibrated on 2014 interval hazards, and evaluated on the common 2015 cohort. Its 12-, 24-, and 36-month ROC-AUC values are 0.676, 0.668, and 0.666; the lower ranking performance is the cost of a more realistic, but still approximate, event-time formulation and provides a useful sensitivity check rather than a new champion model.

7.4 Expected loss translation

The project converts probability into an illustrative credit-loss measure using the standard identity

$$EL_i = PD_i \times LGD_i \times EAD_i.$$

It assumes a constant $LGD = 60\%$ and sets exposure at default equal to the original loan amount. For the resolved 2015 lifetime cohort, this gives \$607.3 million of modelled expected loss on \$5.499 billion of original exposure, or 11.05%.

This translation is valuable because it puts model output in monetary terms. It is also deliberately simplified. Actual EAD changes with amortization and prepayment; LGD varies with recoveries, collateral, costs, and time; and a full accounting or capital calculation would incorporate discounting, scenario weights, and policy definitions.

8 What the results do not yet establish

The project is directionally strong, but several limitations materially affect how its numbers should be used.

1. **Resolved-outcome selection is the main limitation of the lifetime model.** Current and delinquent loans are excluded. This is especially important for 60-month loans issued in 2015, only 67.1% of which have a terminal status in the source snapshot. The resulting default rates are not unbiased vintage rates.
2. **Default timing is approximate.** The horizon labels use `last_pymnt_d`, not a clean charge-off or legal default date. Last payment may precede the actual default event, so the models should be described as approximate early-warning models.
3. **Calibration changes over time and by segment.** The later test vintage has a higher base rate, and underprediction is strongest in high-risk, long-term, and weak-grade groups. Any use of PD in pricing or expected loss needs a recent calibration sample and segment-level checks.
4. **Some predictors embed the lender's existing risk assessment.** Grade, sub-grade, and interest rate are assigned through LendingClub's underwriting. Their inclusion is reasonable for portfolio monitoring, but it can overstate the model's independent underwriting contribution.

The completed ablation shows that borrower and loan fields alone retain useful discrimination (ROC-AUC 0.716), while the full model remains stronger (0.729).

5. **Model interpretability has not yet been demonstrated empirically.** Logistic regression is structurally interpretable, but the notebook does not report coefficients or stability. Grade, sub-grade, and interest rate are also highly related, so individual coefficients may be unstable even when prediction is sound.
6. **The corrected categorical treatment is only one robustness check.** The updated histogram model marks encoded nominal features as categorical. Alternative learners and encodings should still be compared under the same rolling temporal validation design.
7. **Metric uncertainty is not quantified.** The comparison uses one out-of-time year and reports point estimates without bootstrap confidence intervals, vintage-by-vintage backtests, or statistical tests of the small model difference.
8. **The event-time benchmark remains approximate.** The new discrete-time hazard model handles censoring and produces monotone cumulative PD, but it still uses last payment as a default-time proxy and treats prepayment as a competing event censored at payoff. A validated competing-risks design is still needed.
9. **Expected loss is illustrative.** Constant LGD and original loan amount as EAD are transparent assumptions, not validated recovery and exposure models. The results are not IFRS 9, CECL, regulatory-capital, or pricing estimates.
10. **External validity is limited by data vintage.** The file ends in 2018 Q4. Product policy, borrower mix, servicing, and economic conditions may differ materially in a current portfolio.

The raw population counts, status definitions, temporal cohort sizes, and horizon label rates in the notebook were independently reproduced from the supplied source file. Model-performance values in this report come from the corrected notebook's stored outputs. That notebook was executed end to end in a fresh kernel: all 72 code cells completed and no error outputs remained.

9 Working checklist for the next iteration

The project is unfinished. The following checklist keeps future work anchored to the early-warning objective rather than to model complexity for its own sake. The phases are ordered deliberately: later modelling and operating decisions depend on the data and event definitions established first.

Phase 1: Establish the data and target foundation

- ☐ **Audit the available post-origination data.** Confirm whether monthly payment history, days past due, outstanding balance, utilization, bureau updates, hardship events, recoveries, and relevant macroeconomic series are available. Produce a data dictionary, coverage table, and explicit gap list before choosing a dynamic model.
- ☐ **Build a leakage-safe loan-month panel.** Create one observation per loan per reporting month, preserve only information known at that date, and add automated checks for duplicate

periods, impossible dates, outcome leakage, and inconsistent balances. This panel becomes the canonical dataset for subsequent deterioration modelling.

- **Define deterioration and the event clock.** Agree on operational definitions for delinquency migration, material PD increase, Watchlist entry, High Risk entry, default, cure, prepayment, and exit. Replace last payment with the best available event dates and state how competing events will be handled.

Phase 2: Develop a genuinely dynamic risk model

- **Create simple behavioural benchmarks first.** Compare the current origination model with transparent additions such as recent delinquency status, missed-payment counts, payment-to-installment ratio, balance trend, and utilization change. Require each added signal to improve out-of-time warning performance.
- **Fit a monthly event-time model.** Estimate conditional 3-, 6-, and 12-month default risk with a discrete-time hazard or competing-risks model using time-varying features. Compare it with the existing horizon classifiers on the same loan-month cohorts and cutoffs.
- **Model risk-state transitions.** Estimate movement among Healthy, Watchlist, High Risk, Default, Cure, and Exit. Report transition matrices, state persistence, and time from first warning to default rather than evaluating only terminal classification.

Phase 3: Turn scores into stable and explainable alerts

- **Set state rules with persistence and hysteresis.** Define entry, escalation, cure, and exit thresholds using calibrated absolute PD, change in PD, and severe-event overrides. Require persistence where appropriate so small score movements do not repeatedly move an account between states.
- **Measure whether warnings arrive early enough.** Report the share of defaults and loss exposure identified 3, 6, and 12 months in advance, median alert lead time, precision and recall at review capacity, false alerts, and average time spent on the Watchlist.
- **Explain each alert at borrower level.** Show current PD, change from the previous observation, horizon of concern, exposure, expected loss, recent behavioural events, and a small set of stable reason codes. Validate that the explanations remain consistent under correlated predictors.

Phase 4: Validate decisions and prepare for controlled use

- **Run rolling temporal and segment backtests.** Repeat fit–calibrate–test splits over multiple vintages. Track discrimination, calibration, loss capture, warning lead time, and state stability with confidence intervals by term, grade, and material borrower segment.
- **Develop realistic LGD and EAD components.** Estimate recoveries, collection costs, outstanding exposure, amortization, and prepayment so that the economic layer reflects loss at the warning horizon rather than original balance and constant severity.

- **Validate intervention economics.** Replace the illustrative 15% effectiveness, \$50 review cost, and 20% capacity with policy-owned estimates. Where possible, use a retrospective comparison or controlled pilot to test which actions reduce loss, for which accounts, and at what cost.
- **Define monitoring and governance before deployment.** Assign owners for data quality, model changes, thresholds, overrides, and alert outcomes. Set review triggers for drift, calibration failure, subgroup deterioration, capacity breaches, and realized loss, and document the conditions for rollback or model retirement.

10 Questions worth answering next

- How stable is the borrower-and-loan-only model’s discrimination after removing LendingClub grade, sub-grade, and interest rate across rolling vintages and borrower segments?
- Does the 2014 calibration improve later vintages and high-risk segments, or will calibration require a rolling or segment-specific design?
- Which features distinguish emerging-risk loans from loans that are risky immediately, after controlling for term and grade?
- How stable are the watchlist shares and default rates across issue months, vintages, and macroeconomic regimes?
- What review threshold maximizes expected loss avoided after review cost, false positives, and operational capacity are included?
- How much do realistic amortizing EAD and segment-specific LGD change the apparent concentration of loss by term, grade, and watchlist group?

A Detailed result tables

Table 8. Logistic-regression calibration by risk quintile.

Risk band	Loans	Mean predicted PD	Observed default	Gap
Very low	75,110	5.40%	5.49%	0.08 pp
Low	75,109	10.60%	11.05%	0.45 pp
Medium	75,109	15.38%	17.30%	1.92 pp
High	75,109	21.46%	24.77%	3.31 pp
Very high	75,109	34.39%	42.32%	7.93 pp

Table 9. Histogram-gradient-boosting calibration by risk quintile.

Risk band	Loans	Mean predicted PD	Observed default	Gap
Very low	75,110	4.74%	5.24%	0.50 pp
Low	75,109	9.61%	11.09%	1.48 pp
Medium	75,109	14.70%	16.95%	2.26 pp
High	75,109	21.30%	25.08%	3.78 pp
Very high	75,109	35.52%	42.56%	7.03 pp

B Reproducibility note

The report is based on the current corrected notebook `CreditRisk_fixed.ipynb` and the local LendingClub data extract `accepted_2007_to_2018Q4.csv.gz`. Figures in this document are re-drawn from the notebook’s executed numeric outputs so that the LaTeX source remains portable and the charts remain sharp when printed. The corrected notebook is the companion computational artifact for model fitting and row-level analysis; the original `CreditRisk.ipynb` remains unchanged.